

Coordinate-Free Mathematics Through Experiment

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1 Interference, states, and measurement

Why do observed waves interfere, and why do quantum measurements return discrete outcomes?

1.1 Superposition

A **real vector space** is an abelian group V with scalar multiplication by $\mathbb{R}$ satisfying the distributive and associative laws. A map $T : V \rightarrow W$ is **linear** if

$$T(av + bw) = aT(v) + bT(w).$$

The **direct sum** $V \oplus W$ is the product vector space. The **quotient** V/N identifies v and w when $v - w \in N$.

Exercise 1.1 (String vibrations obey superposition) *For a homogeneous string of length L , constant tension T , density ρ , small transverse displacement, and fixed endpoints, derive*

$$\rho \partial_t^2 u = T \partial_x^2 u, \quad u(0, t) = u(L, t) = 0.$$

Prove that the solutions form a vector space and that evolution from initial data is linear. Derive the normal modes, their frequencies, and conservation of

$$E(t) = \frac{1}{2} \int_0^L (\rho |\partial_t u|^2 + T |\partial_x u|^2) dx.$$

State which small-displacement and no-damping assumptions are required.

Exercise 1.2 (Fourier modes produce interference fringes) *Solve the fixed-end wave equation by expanding initial data in the orthogonal functions $\sin(n\pi x/L)$. Prove Parseval's identity for a finite mode sum and use it to express the conserved energy as a sum of modal energies. For two coherent monochromatic amplitudes ψ_1, ψ_2 , derive*

$$|\psi_1 + \psi_2|^2 = |\psi_1|^2 + |\psi_2|^2 + 2 \operatorname{Re}(\overline{\psi_1} \psi_2).$$

Determine when the cross term gives constructive and destructive interference, and compare the predicted fringe spacing with Young's two-source interference geometry [2].

The **tensor product** $V \otimes W$ is a vector space equipped with a bilinear map $(v, w) \mapsto v \otimes w$ such that every bilinear map $V \times W \rightarrow X$ factors through a unique linear map $V \otimes W \rightarrow X$.

Exercise 1.3 (Tensor products model composites) *Construct $V \otimes W$ as a quotient of the free vector space on $V \times W$ and prove its universal property. Prove uniqueness up to unique isomorphism. Explain why closure under superposition forces $V \otimes W$ to contain non-product vectors.*

1.2 Spectral measurement

A **finite-dimensional Hilbert space** is a finite-dimensional complex vector space H with a positive-definite Hermitian inner product. The **adjoint** A^* satisfies $\langle Av, w \rangle = \langle v, A^*w \rangle$. An operator is **self-adjoint** if $A = A^*$ and **normal** if $A^*A = AA^*$. An **orthogonal projection** satisfies $P = P^* = P^2$.

Exercise 1.4 (Spectral measurement) *Prove that every normal $A \in \operatorname{End}(H)$ has a unique decomposition*

$$A = \sum_{\lambda \in \sigma(A)} \lambda P_\lambda, \quad P_\lambda P_\mu = 0 \quad (\lambda \neq \mu), \quad \sum_{\lambda} P_\lambda = 1.$$

For self-adjoint A and a unit state ψ , prove that $\langle \psi, P_\lambda \psi \rangle$ is a probability distribution. For $A = \sigma_3$, compare the two predicted outcomes with the beam splitting in the Stern–Gerlach experiment [1].

Exercise 1.5 (Noncommuting observables are uncertain) For self-adjoint A, B and a unit vector ψ , define

$$\Delta_\psi A = \|(A - \langle \psi, A \psi \rangle) \psi\|.$$

Apply Cauchy–Schwarz to prove

$$\Delta_\psi A \Delta_\psi B \geq \frac{1}{2} |\langle \psi, [A, B] \psi \rangle|.$$

Characterize equality and compute the bound for two Pauli observables. State why the result constrains repeated preparations rather than simultaneous classical values assigned to one particle.

For $\psi \in H_A \otimes H_B$, define $T_\psi : H_A^* \rightarrow H_B$ by $T_\psi(f) = (f \otimes 1)(\psi)$.

Exercise 1.6 (Schmidt spectra measure entanglement) Apply singular-value decomposition to T_ψ and prove

$$\psi = \sum_{i=1}^r s_i a_i \otimes b_i$$

for orthonormal families and $s_i > 0$. Prove that ψ is a product vector exactly when $r = 1$, and identify s_i^2 as the nonzero eigenvalues of both reduced density operators.

1.3 Open evolution

A **state** on $M_n(\mathbb{C})$ is a positive matrix ρ with $\text{tr } \rho = 1$. A linear map $\Phi : M_n \rightarrow M_m$ is **positive** if it preserves positive matrices and **completely positive** if $1_{M_r} \otimes \Phi$ is positive for every r . A **quantum channel** is completely positive and trace preserving. Its **Choi matrix** is

$$C_\Phi = \sum_{i,j} E_{ij} \otimes \Phi(E_{ij}).$$

Exercise 1.7 (Ancillas force complete positivity) Prove that transposition is positive but not 2-positive by applying partial transpose to a maximally entangled two-qubit state. Prove the equivalences

$$\Phi \text{ completely positive} \iff C_\Phi \geq 0 \iff \Phi(x) = \sum_a K_a x K_a^*.$$

Construct a Stinespring dilation from the Kraus operators and compare the Choi state reconstruction with quantum process tomography [3, 4, 5].

Exercise 1.8 (No channel clones nonorthogonal states) Suppose a channel maps $\rho \otimes \sigma$ to $\rho \otimes \rho$ for every pure state ρ . Use contractivity of trace distance, or monotonicity of fidelity, to obtain a contradiction from two nonorthogonal states [6].

2 Molecular spectra and symmetry

Why do experimentally observed spectral lines organize into symmetry types and selection rules?

2.1 Representations and characters

A **representation** of a group G on V is a homomorphism $\rho : G \rightarrow \text{GL}(V)$. A subspace is **invariant** if it is preserved by every $\rho(g)$ and **irreducible** if it has no nontrivial invariant subspaces. An **intertwiner** $T : V \rightarrow W$ satisfies $T\rho_V(g) = \rho_W(g)T$.

Exercise 2.1 (Finite symmetry decomposes into sectors) For finite G , average an inner product and a projection over G . Deduce Maschke's theorem and prove Schur's lemma over $\mathbb{C}$. If a Hamiltonian commutes with G , prove that it is block diagonal on

$$V \cong \bigoplus_{\alpha} M_{\alpha} \otimes V_{\alpha}$$

and identify the symmetry-enforced degeneracy of each block.

The **character** of V is $\chi_V(g) = \text{tr}(\rho(g))$. For class functions define

$$\langle \chi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} \psi(g).$$

Exercise 2.2 (Characters recover multiplicities) Prove the orthogonality of irreducible characters and

$$m_{\alpha} = \langle \chi_{\alpha}, \chi_V \rangle_G.$$

Prove that

$$P_\alpha = \frac{\dim V_\alpha}{|G|} \sum_{g \in G} \overline{\chi_\alpha(g)} \rho(g)$$

projects onto the V_α -isotypic component.

2.2 Water vibrations and selection rules

For a nonlinear molecule with N nuclei, the displacement representation has dimension $3N$. Its vibrational representation is obtained by removing the translation and rotation representations. In the electric-dipole model a transition $V_i \rightarrow V_f$ through a dipole component V_μ is allowed only if

$$\text{Hom}_G(1, V_f^* \otimes V_\mu \otimes V_i) \neq 0.$$

Exercise 2.3 (Water has three symmetry-classified vibrations)

Using the equilibrium point group C_{2v} , compute the character of the nine-dimensional displacement representation of water and prove

$$V_{\text{vib}} \cong 2A_1 \oplus B_2.$$

Determine the infrared- and Raman-active modes. Compare the predicted mode count and symmetry types with measured water spectra [7]. Distinguish the empirical equilibrium geometry and force constants from the representation-theoretic conclusions.

Exercise 2.4 (Symmetry gives electric-dipole selection rules) Let V_i, V_f be irreducible energy sectors and let the dipole operator transform as the spatial vector representation V_μ . Prove that a nonzero transition amplitude is equivalent to the occurrence of the trivial representation in $V_f^* \otimes V_\mu \otimes V_i$. For rotation symmetry, use the decomposition of $V_j \otimes V_1$ to derive $\Delta j = 0, \pm 1$ with the $0 \rightarrow 0$ exception. Separate this symmetry rule from parity, radial overlap, and experimental line strength.

2.3 Continuous and projective symmetry

A **matrix Lie group** is a closed subgroup of $\text{GL}(n, \mathbb{C})$. Its **Lie algebra** consists of the matrices X for which e^{tX} stays in the group, with bracket

$[X, Y] = XY - YX$. For rotation generators use

$$[J_i, J_j] = i\varepsilon_{ijk}J_k, \quad J^2 = J_1^2 + J_2^2 + J_3^2.$$

Exercise 2.5 (Rotation symmetry produces spin) *Prove that J^2 commutes with every J_i . Using $J_{\pm} = J_1 \pm iJ_2$, classify the finite-dimensional unitary irreducible representations by $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ and $m = -j, -j+1, \dots, j$. Verify the spin-1/2 representation with Pauli matrices and relate its two outcomes to Stern–Gerlach splitting.*

Exercise 2.6 (Hydrogen’s hidden rotational symmetry) *For the Coulomb Hamiltonian, define angular momentum $L = q \times p$ and the Runge–Lenz vector in the bound-state model. Prove that their commutators close to $\mathfrak{so}(4)$ after rescaling by the negative energy. Use the irreducible representation dimensions to recover the n^2 orbital degeneracy. Identify relativistic, spin, and radiative effects omitted by the model and compare the resulting splitting hierarchy with hydrogen spectroscopy [8].*

A **projective representation** satisfies

$$U_g U_h = \alpha(g, h) U_{gh},$$

where associativity makes α a $U(1)$ -valued 2-cocycle. Rephasing U_g changes α by a coboundary.

Exercise 2.7 (Cohomology classifies projective symmetry) *Prove that equivalence classes of factor systems are classified by $H^2(G, U(1))$. For $G = \mathbb{Z}/2 \times \mathbb{Z}/2$ and $U_x = \sigma_1$, $U_z = \sigma_3$, compute the nontrivial class and prove that every commuting edge operator is scalar. State the symmetry and gap assumptions required for protection of a spin-1/2 edge doublet.*

3 Electromagnetic fields and geometry

Why do Maxwell fields, charge conservation, and classical motion admit coordinate-independent formulations?

3.1 Forms and Maxwell equations

For a smooth manifold M , a **differential p -form** is a smooth section of $\Lambda^p T^*M$. The exterior derivative $d : \Omega^p(M) \rightarrow \Omega^{p+1}(M)$ satisfies $d^2 = 0$ and the graded Leibniz rule. For an oriented manifold with boundary, Stokes' theorem is

$$\int_M d\omega = \int_{\partial M} \omega.$$

A pseudo-Riemannian metric and orientation determine the **Hodge star** by

$$\alpha \wedge \star \beta = \langle \alpha, \beta \rangle_g \text{vol}_g.$$

On four-dimensional spacetime, the electromagnetic field is $F \in \Omega^2(M)$ and the current is $J \in \Omega^3(M)$.

Exercise 3.1 (Maxwell equations force charge conservation) *From*

$$dF = 0, \quad d \star F = J,$$

prove $dJ = 0$ and $\int_{\partial X} J = 0$ for every compact four-region X . Recover the coordinate continuity equation and identify it as a chart expression of the invariant statement. Derive the vacuum wave equation and compare its wave speed with electromagnetic-wave measurements [9].

3.2 Hamiltonian mechanics

A **symplectic form** is a closed nondegenerate two-form ω . A Hamiltonian H determines X_H by

$$\iota_{X_H} \omega = dH.$$

The **Poisson bracket** is $\{f, h\} = \omega(X_f, X_h)$.

Exercise 3.2 (Hamiltonian flow preserves phase volume) *Using Cartan's formula, prove*

$$\mathcal{L}_{X_H} \omega = 0$$

and deduce preservation of $\omega^n/n!$. Derive Hamilton's equations in canonical coordinates. Explain why friction cannot be an autonomous Hamiltonian flow on the same finite-dimensional phase space.

Exercise 3.3 (Magnetic curvature produces the Lorentz force) *On T^*Q , replace the canonical symplectic form by $\omega_B = \omega_0 + q\pi^*B$ for a closed two-form B . Derive the Lorentz force from the kinetic Hamiltonian and identify gauge change as a change of potential leaving B fixed.*

3.3 Metric curvature

A **pseudo-Riemannian metric** is a smooth nondegenerate symmetric bilinear form on TM . Its **Levi-Civita connection** is torsion free and metric compatible. Its curvature is

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Exercise 3.4 (Curvature produces interferometer strain) *Derive the coordinate-free Koszul formula and prove existence and uniqueness of the Levi-Civita connection. For a variation of geodesics with tangent T and separation J , prove*

$$\nabla_T \nabla_T J = R(T, J)T.$$

Apply this equation to a weak transverse plane gravitational wave and derive the opposite fractional arm-length changes measured by an equal-arm Michelson interferometer. Compare the predicted differential strain with GW150914 [10] and identify the weak-field and freely falling mirror assumptions.

3.4 Connections and gauge fields

A **vector bundle** $E \rightarrow M$ is locally a product $U \times V$ with linear transition maps. A **connection** is a linear map

$$\nabla : \Omega^0(M, E) \rightarrow \Omega^1(M, E), \quad \nabla(fs) = df \otimes s + f\nabla s.$$

Its **curvature** is $F_\nabla = \nabla^2 \in \Omega^2(M, \text{End } E)$. In a local gauge, $F_A = dA + A \wedge A$.

Exercise 3.5 (Curvature is gauge covariant) *For $A^g = g^{-1}Ag + g^{-1}dg$, prove*

$$F_{A^g} = g^{-1}F_A g, \quad d_A F_A = 0.$$

Using an invariant inner product, vary

$$S(A) = \frac{1}{2} \int_M \langle F_A \wedge \star F_A \rangle$$

and derive the Yang–Mills equation. Recover Maxwell theory for $G = U(1)$ and distinguish the empirical gauge group and matter representations from the geometric consequences.

4 Observed phase and quantized transport

How can a field-free region carry an observable phase, and why is Hall conductance quantized?

4.1 Aharonov–Bohm holonomy

The **de Rham cohomology** is

$$H_{\text{dR}}^p(M) = \ker(d : \Omega^p(M) \rightarrow \Omega^{p+1}(M)) / \text{im}(d : \Omega^{p-1}(M) \rightarrow \Omega^p(M)).$$

For a $U(1)$ connection A and closed loop γ , define

$$\text{Hol}_A(\gamma) = \exp\left(\frac{iq}{\hbar} \int_\gamma A\right).$$

Exercise 4.1 (A field-free loop carries phase) Outside an ideal solenoid let

$$M = \mathbb{R}^3 \setminus \{(0, 0, z) : z \in \mathbb{R}\}, \quad A = \frac{\Phi}{2\pi} d\theta.$$

Prove that the phase depends only on $[A] \in H_{\text{dR}}^1(M)$ and $[\gamma] \in H_1(M)$, and equals $\exp(iq\Phi/\hbar)$ for a once-winding loop. Compare the predicted fringe shift with shielded-flux electron interferometry [11, 12].

Exercise 4.2 (Berry phase is eigenbundle holonomy) For a smooth family of self-adjoint operators with a simple isolated eigenvalue, prove that the eigenlines form a bundle. For a local unit eigenvector ψ , set

$$\mathcal{A} = i\langle \psi, d\psi \rangle, \quad \mathcal{F} = d\mathcal{A}.$$

Prove the gauge-transformation law and identify $\exp(i \int_\gamma \mathcal{A})$ with holonomy. For $H(n) = n \cdot \sigma$, $n \in S^2$, compute the phase as half the enclosed solid angle and compare it with neutron spin-rotation measurements [13].

4.2 Chern classes and stable bundles

For a complex bundle E with connection ∇ , define the Chern forms by

$$\det\left(1 + \frac{iF_\nabla}{2\pi}\right) = 1 + c_1(E, \nabla) + c_2(E, \nabla) + \cdots.$$

The **Grothendieck group** $K^0(X)$ is generated by vector bundles with $[E \oplus F] = [E] + [F]$.

Exercise 4.3 (Curvature produces stable characteristic classes)

Using the Bianchi identity, prove that the Chern forms are closed. For a path of connections, derive the transgression formula and prove that their de Rham classes are connection independent. Prove that

$$\text{ch} : K^0(X) \longrightarrow H_{\text{dR}}^{\text{even}}(X)$$

is well defined and that a nonzero first Chern number obstructs a global frame.

4.3 Integer quantum Hall effect

For a smooth gapped self-adjoint family $H(k)$ on the Brillouin torus, define the occupied projection by

$$P(k) = \frac{1}{2\pi i} \int_{\Gamma} (z - H(k))^{-1} dz.$$

Its images form the **Bloch bundle** E_{occ} .

Exercise 4.4 (Gap-preserving Hall integer) *Prove that a gap-preserving homotopy produces isomorphic endpoint Bloch bundles and preserves*

$$C_1(P) = \frac{i}{2\pi} \int_{\mathbb{T}^2} \text{tr}(P dP \wedge dP) \in \mathbb{Z}.$$

Assuming the TKNN formula $\sigma_{xy} = C_1(P)e^2/h$, compare the predicted plateaus with the integer quantum Hall measurements [14, 15]. Identify gap closing as the point where the invariant may change.

4.4 Fredholm index

A bounded operator $T : H_0 \rightarrow H_1$ is **Fredholm** if its kernel and cokernel are finite dimensional and its range is closed. Define

$$\text{ind } T = \dim \ker T - \dim \text{coker } T.$$

Exercise 4.5 (Zero-mode imbalance is stable) *Prove that T is Fredholm exactly when its image in the Calkin algebra is invertible. Deduce local constancy and invariance under compact perturbation of the index. State the Atiyah–Singer identification of analytic index with the K -theoretic symbol pairing and derive the de Rham Euler-characteristic case [16].*

Exercise 4.6 (Instanton charge produces chiral zero modes) *For an $SU(N)$ connection on a bundle over S^4 , define*

$$Q(A) = \frac{1}{8\pi^2} \int_{S^4} \text{tr}(F_A \wedge F_A).$$

Use Chern–Weil theory to prove $Q(A) \in \mathbb{Z}$. Assuming the coupled Dirac case of the index theorem, relate $Q(A)$ to the difference of chiral zero-mode dimensions and derive the integrated anomaly for N_f massless fermions. Distinguish the mathematical index identity from the physical assumptions in the fermion model.

5 Bell experiments and noncommutative analysis

Why do entangled systems violate classical correlation bounds while remaining unable to signal?

5.1 Matrix norms and distinguishability

A **concrete operator space** is a closed subspace $E \subseteq \mathcal{B}(H, K)$ with matrix norms inherited from

$$M_r(E) \subseteq \mathcal{B}(H^{\oplus r}, K^{\oplus r}).$$

For $\phi : E \rightarrow F$, define

$$\|\phi\|_{\text{cb}} = \sup_r \|1_{M_r} \otimes \phi\|.$$

Exercise 5.1 (Matrix levels detect noncommutativity) *Prove for transposition $t : M_n \rightarrow M_n$ that*

$$\|t\| = 1, \quad \|t\|_{\text{cb}} = n.$$

For the row and column spaces R_n and C_n , prove that the identity between their underlying Hilbert spaces has cb norm $\sqrt{n}$ in either direction. Explain why an ordinary Banach norm forgets both distinctions.

For $u = \sum_k x_k \otimes y_k \in E \otimes F$, define the **Haagerup norm**

$$\|u\|_h = \inf \| [x_1 \ \cdots \ x_r] \|_{M_{1,r}(E)} \| [y_1 \ \cdots \ y_r]^T \|_{M_{r,1}(F)}.$$

Exercise 5.2 (Trace norm governs state discrimination) *Prove*

$$C_n \otimes_h R_n \cong M_n, \quad R_n \otimes_h C_n \cong S_1^n.$$

For equally likely states ρ_0, ρ_1 , prove the Helstrom formula

$$p_{\text{succ}} = \frac{1}{2} + \frac{1}{4} \|\rho_0 - \rho_1\|_1.$$

Interpret the trace norm through the second Haagerup product.

For $\Psi : M_n \rightarrow M_m$, define the **diamond norm**

$$\|\Psi\|_{\diamond} = \sup_r \|\Psi \otimes 1_{M_r}\|_{1 \rightarrow 1}.$$

Exercise 5.3 (Entangled probes distinguish channels) *Prove stabilization at ancillary dimension n and*

$$p_{\text{succ}}(\Phi_0, \Phi_1) = \frac{1}{2} + \frac{1}{4} \|\Phi_0 - \Phi_1\|_{\diamond}.$$

Identify the role of Schmidt rank and compare ancilla-assisted discrimination with process-discrimination experiments [17, 18].

An **operator system** is a self-adjoint unital subspace of $\mathcal{B}(H)$. A unital map between operator systems is completely positive when it preserves the positive cone at every matrix level.

Exercise 5.4 (Operator-system positivity) For an operator space $E \subseteq \mathcal{B}(H, K)$, form the Paulsen system

$$\mathcal{S}(E) = \left\{ \begin{pmatrix} \lambda 1_K & x \\ y^* & \mu 1_H \end{pmatrix} : x, y \in E, \lambda, \mu \in \mathbb{C} \right\}.$$

Prove that a map $\phi : E \rightarrow F$ is a complete contraction exactly when its unital extension $\mathcal{S}(E) \rightarrow \mathcal{S}(F)$ is completely positive. Explain why this gives a common language for channel admissibility and matrix-level distinguishability.

5.2 Bell correlations

For self-adjoint unitaries A_0, A_1, B_0, B_1 , define

$$S = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

Exercise 5.5 (Quantum correlations reach the Tsirelson bound)

Prove that deterministic local sign assignments give $|S| \leq 2$. Using

$$S^2 = 4 - [A_0, A_1] \otimes [B_0, B_1],$$

prove $\|S\| \leq 2\sqrt{2}$ and attain equality with a maximally entangled two-qubit state. Compare the quantum prediction with loophole-controlled Bell experiments [19, 20]. Explain why violation excludes the specified local hidden-variable model but does not permit signaling.

Exercise 5.6 (Grothendieck bounds quantum advantage) For real coefficients c_{ij} , compare

$$\sup_{\varepsilon_i, \delta_j \in \{\pm 1\}} \left| \sum_{ij} c_{ij} \varepsilon_i \delta_j \right| \quad \text{and} \quad \sup_{\|x_i\|, \|y_j\| \leq 1} \left| \sum_{ij} c_{ij} \langle x_i, y_j \rangle \right|.$$

Use Tsirelson's vector representation and the real Grothendieck inequality to obtain a dimension-independent bound on bipartite correlation advantage. Identify CHSH as a particular coefficient matrix.

5.3 Infinite observables and index

A **von Neumann algebra** is a unital $*$ -subalgebra $M \subseteq \mathcal{B}(H)$ equal to its bicommutant. For a finite-factor inclusion $N \subseteq M$, let $E_N : M \rightarrow N$ be the trace-preserving conditional expectation and $e_N : L^2(M) \rightarrow L^2(N)$ the Jones projection.

Exercise 5.7 (Weak limits produce the bicommutant) *Prove the finite-dimensional bicommutant theorem directly. State the von Neumann bicommutant theorem for a unital $*$ -subalgebra of $\mathcal{B}(H)$ and show that vector functionals are normal on its weak closure. Explain why weak closure, rather than norm closure alone, retains limits of increasing families of quantum events.*

Exercise 5.8 (Subfactor index produces braid relations) *Prove $e_N x e_N = E_N(x) e_N$. In the basic-construction tower, derive*

$$e_i^2 = e_i = e_i^*, \quad e_i e_{i \pm 1} e_i = \delta^{-2} e_i, \quad e_i e_j = e_j e_i \quad (|i - j| \geq 2).$$

Relate δ^2 to the Jones index and derive braid generators from the Temperley–Lieb relations. State the quantization of subfactor indices below four [21].

6 Observed anyonic braiding

Why can exchanging experimentally realized quasiparticles in two dimensions produce phases beyond bosonic and fermionic signs?

6.1 Braids and fusion

The **braid group** B_n has generators $\sigma_1, \dots, \sigma_{n-1}$ and relations

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, \quad \sigma_i \sigma_j = \sigma_j \sigma_i \quad (|i - j| \geq 2).$$

A **$\mathbb{C}$ -linear monoidal category** has a bilinear tensor product, unit, and coherent associator. A **braiding** is a natural family $c_{x,y} : x \otimes y \rightarrow y \otimes x$ satisfying the hexagon identities. In a semisimple rigid category, decompositions

$$x \otimes y \cong \bigoplus_z N_{xy}^z z$$

are **fusion rules**; $\text{Hom}(z, x \otimes y)$ is a fusion space.

Exercise 6.1 (Braiding represents particle exchange) *Prove that*

$$\rho(\sigma_i) = 1^{\otimes(i-1)} \otimes c_{x,x} \otimes 1^{\otimes(n-i-1)}$$

defines a representation of B_n on $x^{\otimes n}$. Identify the Yang–Baxter equation with a hexagon consequence and prove that imposing $\sigma_i^2 = 1$ gives the symmetric group.

Exercise 6.2 (A three-cocycle controls reassociation) *For G -graded vector spaces, multiply the associator on degrees g, h, k by $\omega(g, h, k) \in U(1)$. Prove that the pentagon identity is the 3-cocycle equation and that changing ω by a coboundary gives a monoidally equivalent category. For $G = \mathbb{Z}/2$, compute the two classes in $H^3(G, U(1))$ and their pointed fusion rules.*

The **monodromy** is $M_{x,y} = c_{y,x}c_{x,y}$.

Exercise 6.3 (One-third anyons produce a $2\pi/3$ phase) *Assume all simple objects are invertible and form $\mathbb{Z}/m$, generated by a . Prove that scalar monodromy compatible with fusion has the form*

$$M_{a^r, a^s} = \exp\left(\frac{2\pi i k r s}{m}\right).$$

For $m = 3$, $k = 1$, derive the phase advance per enclosed anyon and compare it with interferometric evidence for anyonic braiding [22]. State the ideal topological assumptions and the non-topological device effects that must be controlled.

Exercise 6.4 (Fusion dimension) *Let positive numbers d_x satisfy $d_x d_y = \sum_z N_{xy}^z d_z$. Prove that d is a positive character of the fusion ring. For Fibonacci fusion $\tau \otimes \tau = 1 \oplus \tau$, compute d_τ and the asymptotic growth of the fusion space of n τ -anyons. Relate nonintegral dimension to state growth rather than to a fractional dimension of an ordinary vector space.*

6.2 Chern–Simons theory

For a connection A on an oriented three-manifold, define

$$S_{\text{CS}}(A) = \frac{k}{4\pi} \int_M \text{tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right).$$

A Wilson loop is $W_R(K) = \text{tr}_R \text{Hol}_A(K)$.

Exercise 6.5 (Chern–Simons link invariants) *Vary S_{CS} and derive $F_A = 0$. Show that a large gauge transformation changes the action by $2\pi k$ times an integer and deduce level quantization. Assuming the ribbon-category evaluation rules, prove that closed braid amplitudes are categorical traces and recover the relation to Jones-type invariants [23, 24].*

6.3 Stable functorial invariants

An additive strong monoidal functor preserves direct sums and tensor products. The **Grothendieck group** $K_0(\mathcal{C})$ is generated by objects with $[x \oplus y] = [x] + [y]$ and multiplication $[x][y] = [x \otimes y]$.

Exercise 6.6 (Physical structures produce stable observables)

Prove that an additive strong monoidal functor induces a ring homomorphism on K_0 and preserves categorical traces when it preserves duals. Organize the Aharonov–Bohm phase, Hall integer, Fredholm index, and anyon monodromy as

$$\text{physical structure} \longrightarrow \text{stable class} \longrightarrow \text{pairing or trace} \longrightarrow \text{observable}.$$

For each, identify the experimental observable, the mathematical invariant, and the hypothesis whose failure permits the value to change.

Exercise 6.7 (Six experiments force six successive structures) *For each of the following observations, identify the minimal mathematical object, the central invariant or theorem, and the model assumption tested by the observation:*

1. *two-source interference and linear superposition;*

2. *water vibrational spectra and character multiplicities;*
3. *electromagnetic-wave propagation and $d^2 = 0$;*
4. *Aharonov–Bohm interference and flat holonomy;*
5. *integer Hall plateaus and the first Chern number;*
6. *Bell violation and matrix-level noncommutativity;*
7. *fractional-quantum-Hall interferometry and braid monodromy.*

For every step, exhibit a phenomenon that the preceding structure cannot distinguish. Prove that the resulting progression is

$$\begin{aligned} & \text{linear spaces} \longrightarrow \text{representations} \longrightarrow \text{geometry} \\ & \longrightarrow \text{stable topology} \longrightarrow \text{operator spaces} \longrightarrow \text{braided categories.} \end{aligned}$$

Separate observations from physical modeling assumptions and both from the mathematical conclusions proved in the exercises.

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